



ECE 484: Principles of Safe Autonomy

Lecture 9: Control (part 2) PID control, linear systems

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Slides adapted in part from materials by Prof. Sayan Mitra.

Review: the control problem

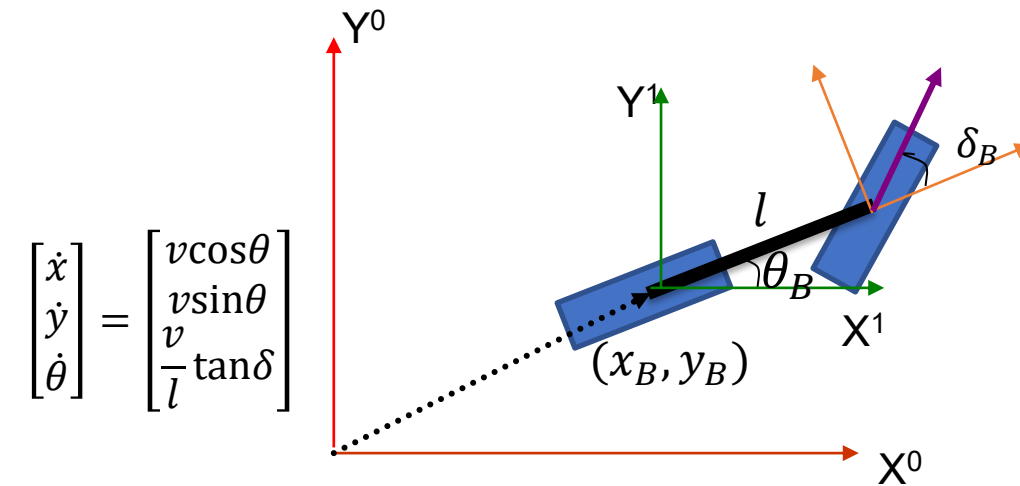
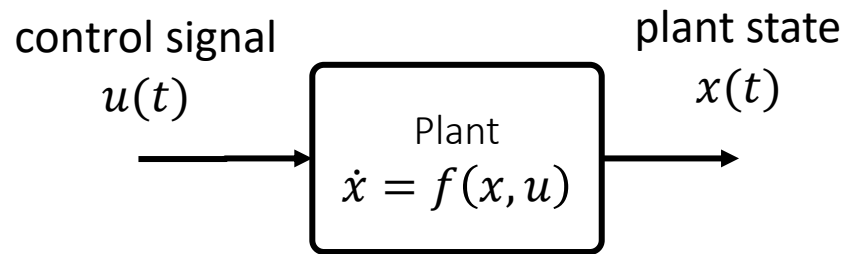
In continuous time, the **dynamics function** defines how the plant state changes continuously with time

Similarly, $u: [0, \infty) \rightarrow \mathbb{R}^m$ models the input signal we control

Given $x(\cdot)$ and any t , $x(t)$ denotes the state of the system at time t

The **Ordinary Differential Equation (ODE)** relates the input and the state $\frac{dx(t)}{dt} = f(x(t), u(t))$

This is written in short as $\frac{dx}{dt} = f(x, u)$ or $\dot{x} = f(x, u)$



bicycle / Dubins car

Outline

Last lecture: Modeling the control problem

- Differential equations; solutions and their properties
- Bang-bang control

This lecture:

- Better controller design (PID control)
- Linear systems

Next lecture:

- Stability Requirements
- Lyapunov theory and its relation to invariance



On-off (bang-bang) control of a room heater with a thermostat

Room: $\dot{x}(t) = u(t) + d(t)$

Heater on or off: $u(t) \in \{0, u_{on}\}$, chosen by

A simple thermostat controller

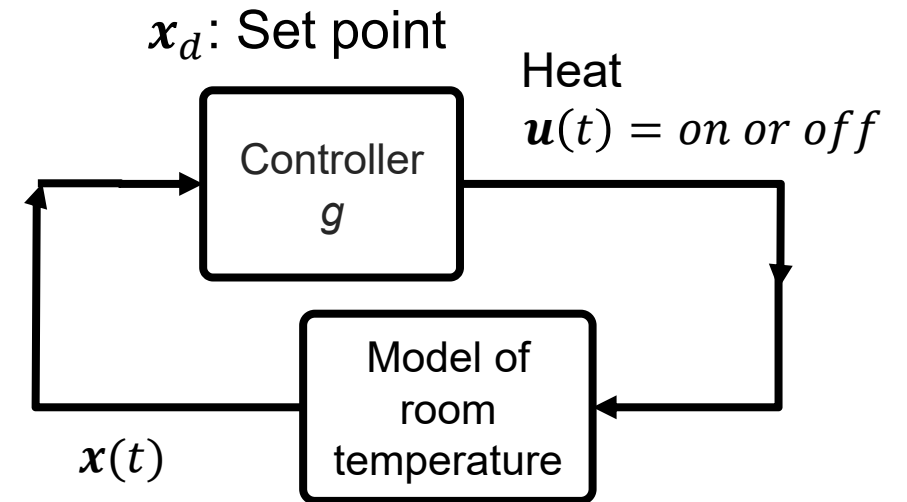
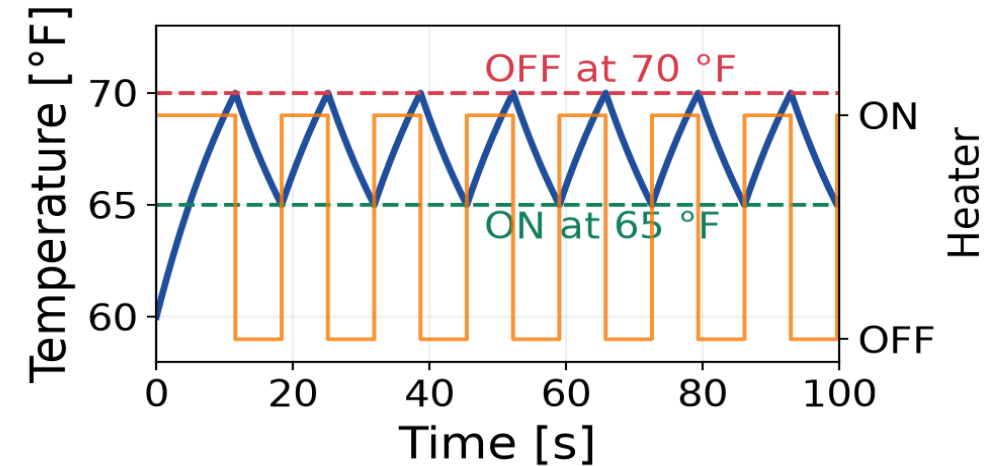
$g(x(t))$:

if $x(t) \geq x_d$ then $u(t) = \text{off}$

else if $x(t) \leq x_d - \varepsilon$ then $u(t) = \text{on}$

otherwise keep the previous u (**hysteresis**: one bit of memory, so $u = g(x, \text{mode})$)

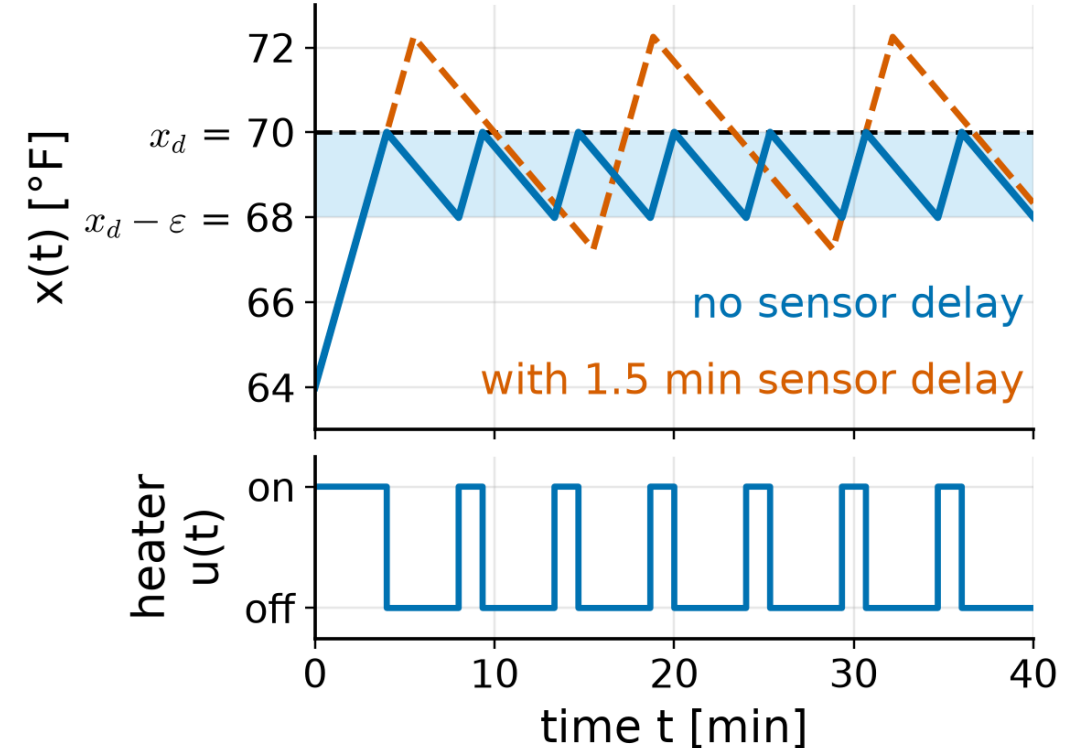
This is called bang-bang control



On-off (bang-bang) control of a room heater with a thermostat

Disadvantages / trade-offs

- Energy efficiency depends on the plant and switching losses.
- Inertia and delays can cause overshoot and undershoot.
- Frequent switching can increase actuator wear.
- Poorly chosen switching or delays can cause instability; bounded cycling alone is not instability.



A Proportional controller

Plant: $\dot{x}(t) = u(t) + d(t)$; d is a disturbance.

Goal: drive the state toward a target (e.g., $x_d = 70$ °F.)

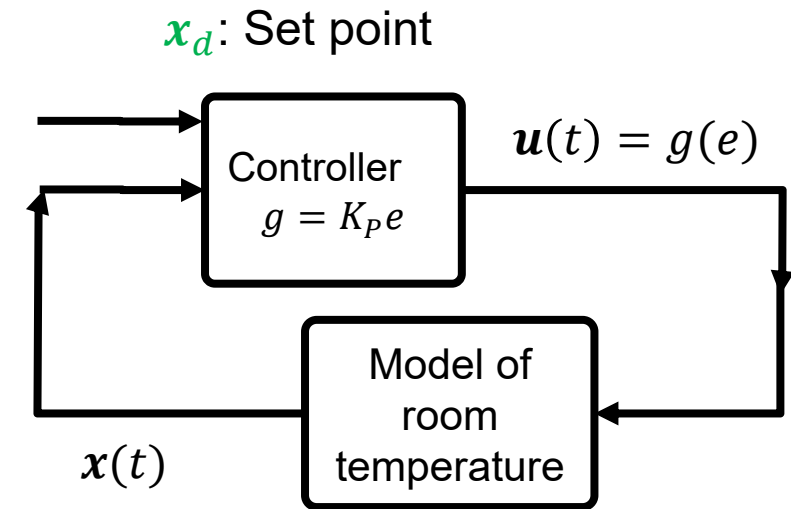
Idea: input opposes deviations from the target (negative feedback).

Error: $e(t) = x_d - x(t)$

Proportional controller: $u(t) = K_P e(t)$; gain $K_P > 0$.

Substitute the feedback law:

$$u(t) = K_P(x_d - x(t))$$
$$\dot{x}(t) = -K_P x(t) + K_P x_d + d(t)$$



A proportional controller: steady state

For the same plant: $\dot{x}(t) = -K_P x(t) + K_P x_d + d(t)$

Simplification: Constant disturbance d_{ss} (e.g., room heat loss):

$$\dot{x}(t) = -K_P x(t) + K_P x_d + d_{ss}$$

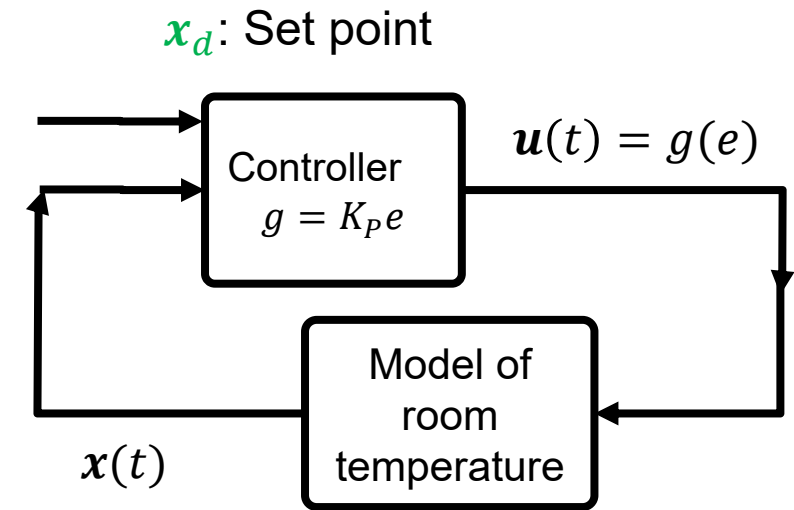
Steady state x_{ss} : set the right-hand side to zero.

$$-K_P x_{ss} + K_P x_d + d_{ss} = 0$$

$$K_P x_{ss} = K_P x_d + d_{ss}$$

$$x_{ss} = x_d + \frac{d_{ss}}{K_P}$$

The problem: x_{ss} does not reach x_d !



Proportional controller example

With constant disturbance d_{ss} we rewrite the ODE

$$\dot{x}(t) = -K_P x(t) + K_P x_d + d_{ss} \text{ with } x_{ss} = \frac{d_{ss}}{K_P} + x_d$$

Rewrite:

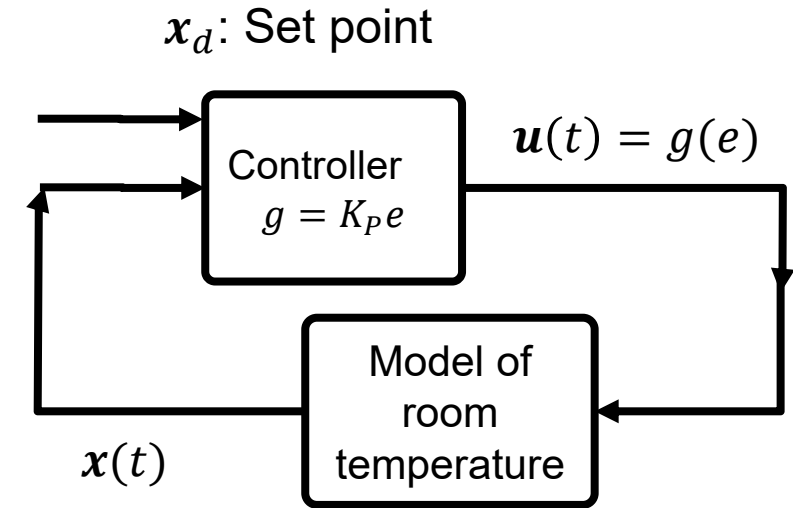
$$\dot{x}(t) = K_P (x_{ss} - x(t))$$

The solution of this ODE (Transient behavior) is:

$$x(t) = x_{ss} + (x(0) - x_{ss})e^{-tK_P}$$

Rewrite:

$$x(t) = x(0)e^{-tK_P} + x_{ss}(1 - e^{-tK_P})$$



General solution of first-order linear DE

$$x(t) = x_{ss} + C e^{-K_P t}$$

Setting $t=0$

$$x(0) = x_{ss} + C$$

Proportional Controller

Transient behavior of the control system

$$x(t) = x(0)e^{-K_P t} + x_{ss}(1 - e^{-K_P t}); x_{ss} = x_d + \frac{d_{ss}}{K_P}$$

P negative feedback tracks the target smoothly in this first-order model.

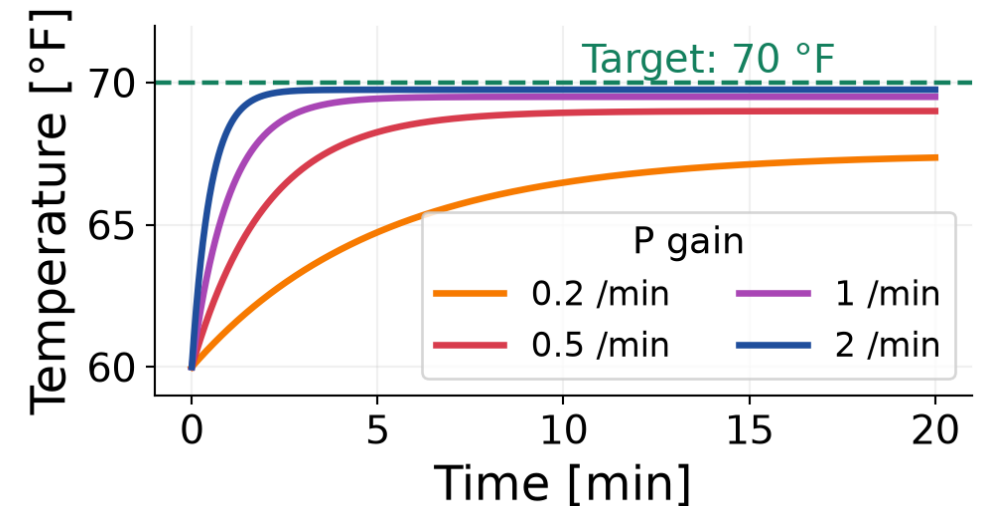
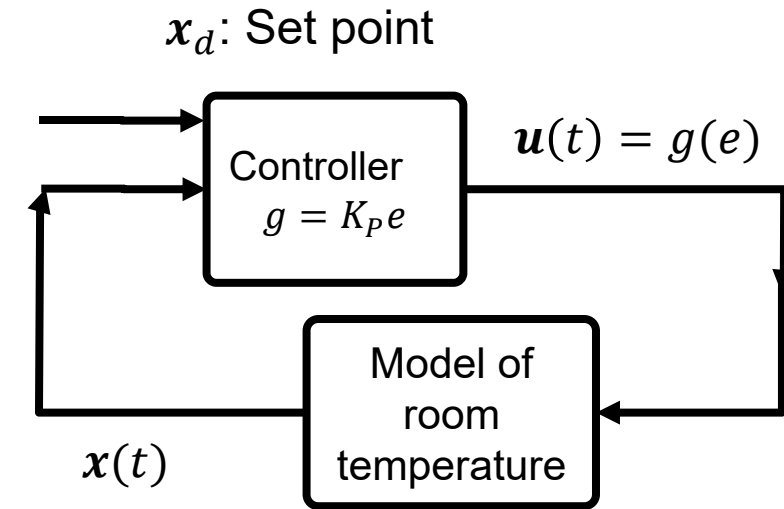
Steady-state error may be nonzero.

Larger K_P gives faster convergence to x_{ss} (this model).

Larger K_P reduces the offset for fixed d_{ss} .

Why not using a super large K_P ?

- With delays or inertia, high gain can cause oscillations or overshoot.
- Actuator limits may make the requested command infeasible (e.g., heater power limit)



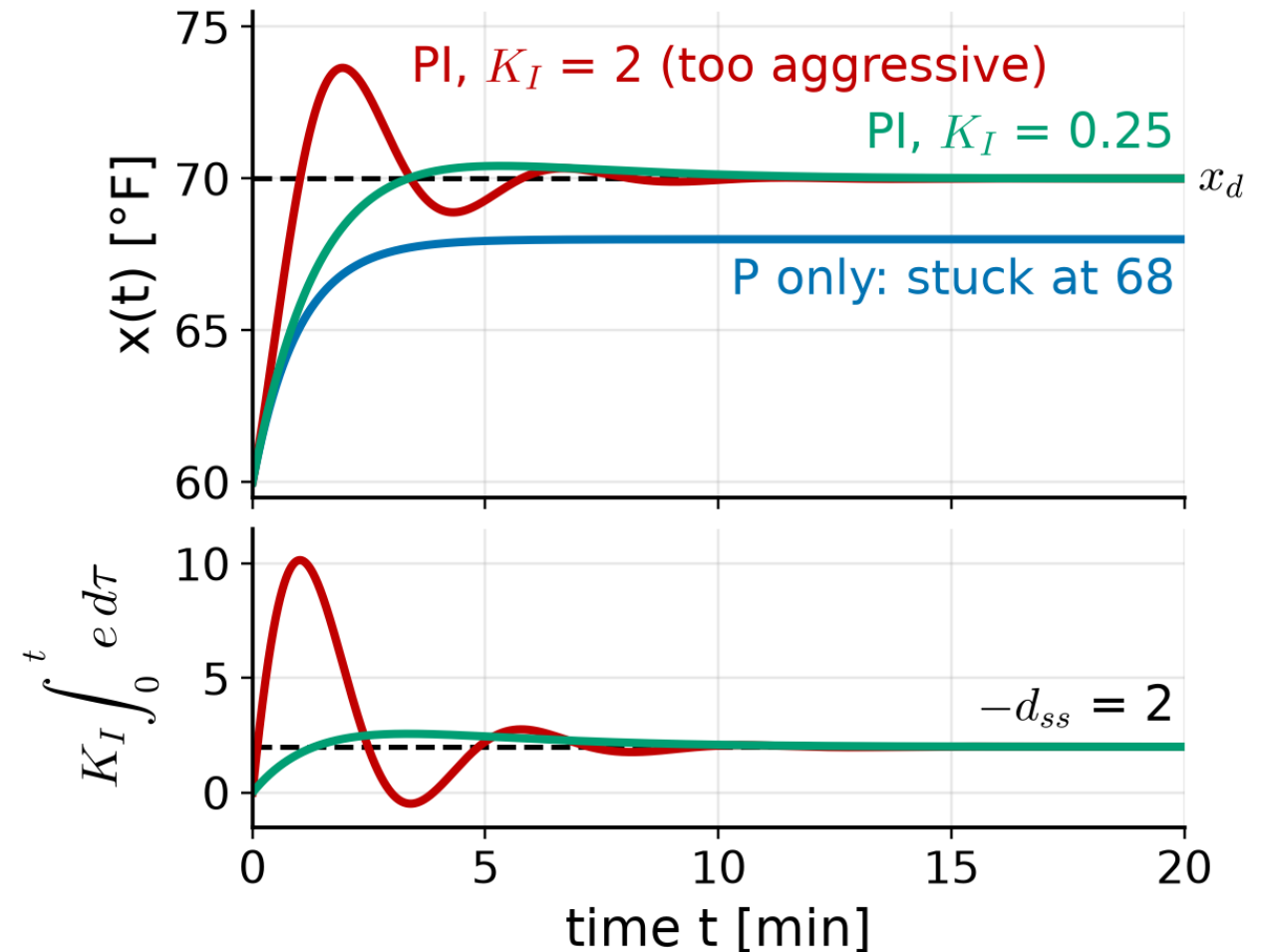
Integral action removes the steady-state error

With P alone the error cannot vanish; add a term that keeps growing as long as any error remains:

$$u(t) = K_P e(t) + K_I \int_0^t e(\tau) d\tau$$

It can only stop growing when $e = 0$. By then it has settled at exactly $-d_{ss}$: the integrator has "learned" the heat loss

Too much K_I : overshoot and ringing



The PID controller

Error: $e(t) = x_d - x(t)$

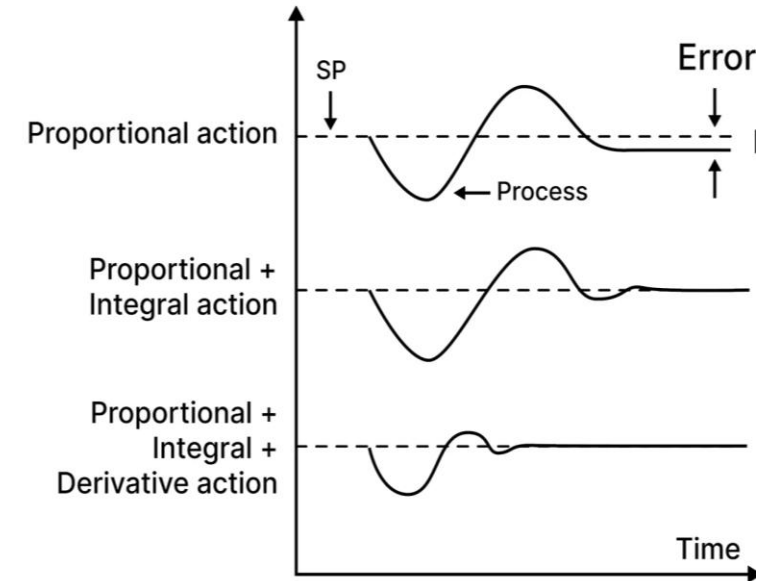
$$u(t) = K_P e(t) + K_I \int_0^t e(\tau) d\tau + K_D \frac{de(t)}{dt}$$

P — present: reacts to the current error. Alone: a steady-state error remains

I — past: accumulates the error ("memory"). Removes the steady-state error; too much causes overshoot

D — future: reacts to the trend of the error. Damps oscillations; amplifies sensor noise

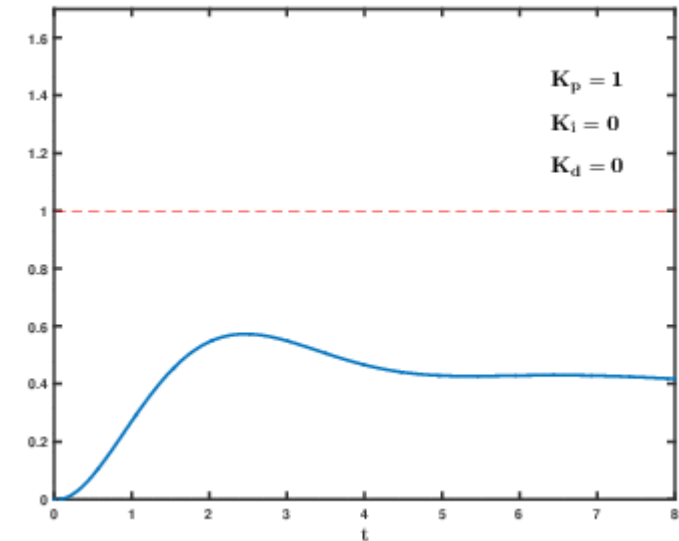
PI control: $K_D = 0$. PD control: $K_I = 0$



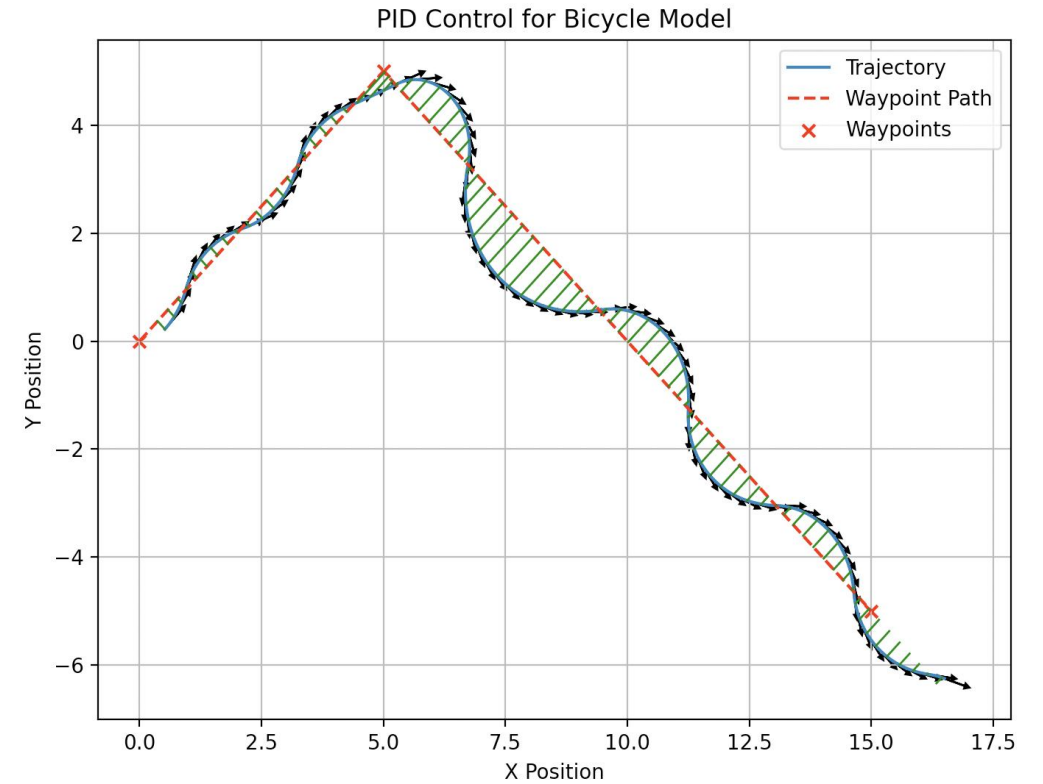
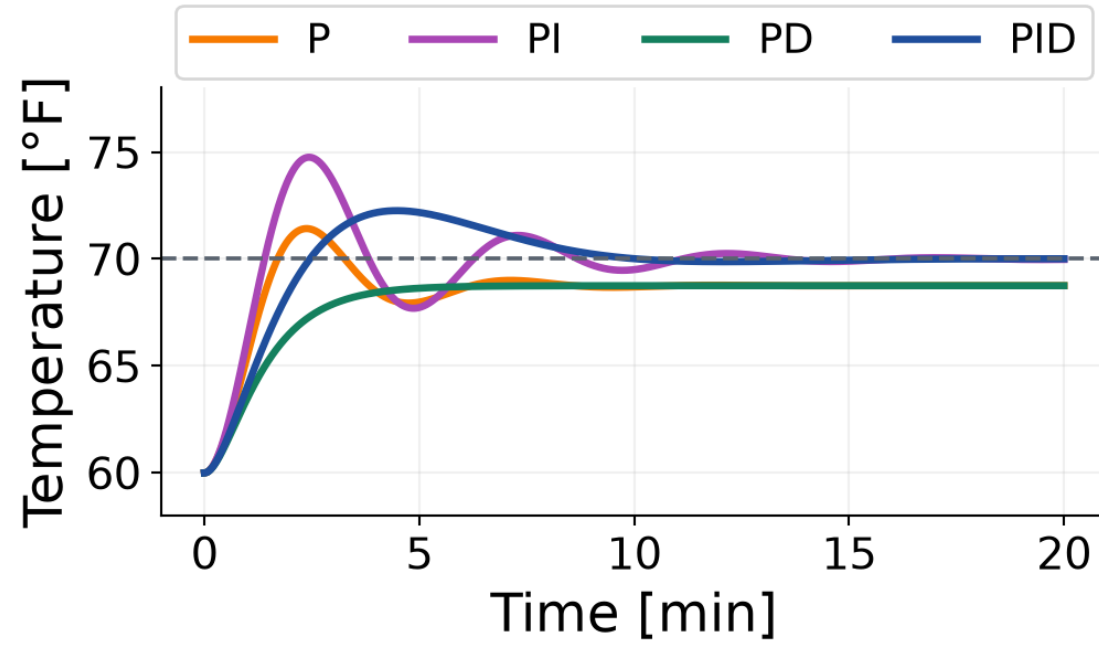
<https://www.realpars.com/blog/pid-vs-advanced-control-methods>

Tuning a PID controller

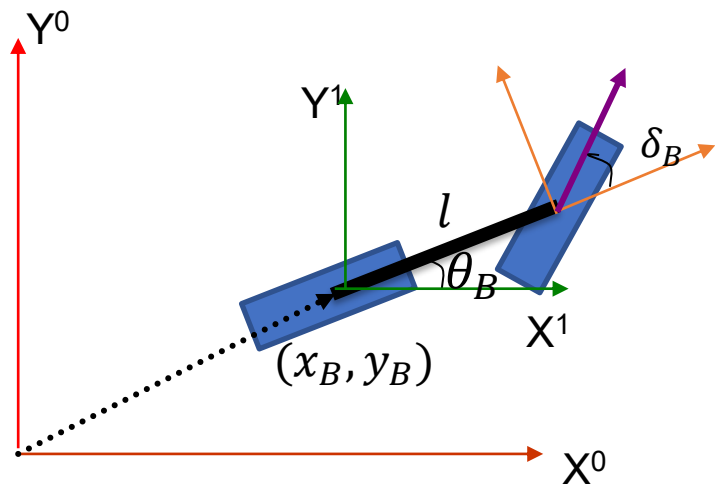
1. Start with $K_I = K_D = 0$. Raise K_P until the response is fast but starts to oscillate; back off
2. Steady-state error left? Add a little K_I . Too much brings the overshoot back
3. Oscillation or overshoot? Add K_D (damping).
Noisy sensor: low-pass filter the derivative
4. Limit the command to what the actuator can do, and stop integrating while it is saturated (anti-windup)



PID controller tuning



Controller design for Rear Wheel Model (Bicycle model)



Plant state (rear-wheel pose): $\mathbf{x}_B = \begin{bmatrix} x_B \\ y_B \\ \theta_B \end{bmatrix} \in \mathbb{R}^3$

Control input: steering angle $u = \delta_B \in \mathbb{R}$

Parameters: wheelbase l ; rear-wheel speed v_B

$$f: \mathbb{R}^3 \times (-\pi/2, \pi/2) \rightarrow \mathbb{R}^3$$

$$\dot{\mathbf{x}}_B = f(\mathbf{x}_B, u)$$

$$\begin{bmatrix} \dot{x}_B \\ \dot{y}_B \\ \dot{\theta}_B \end{bmatrix} = \begin{bmatrix} v_B \cos \theta_B \\ v_B \sin \theta_B \\ \frac{v_B}{l} \tan \delta_B \end{bmatrix}$$

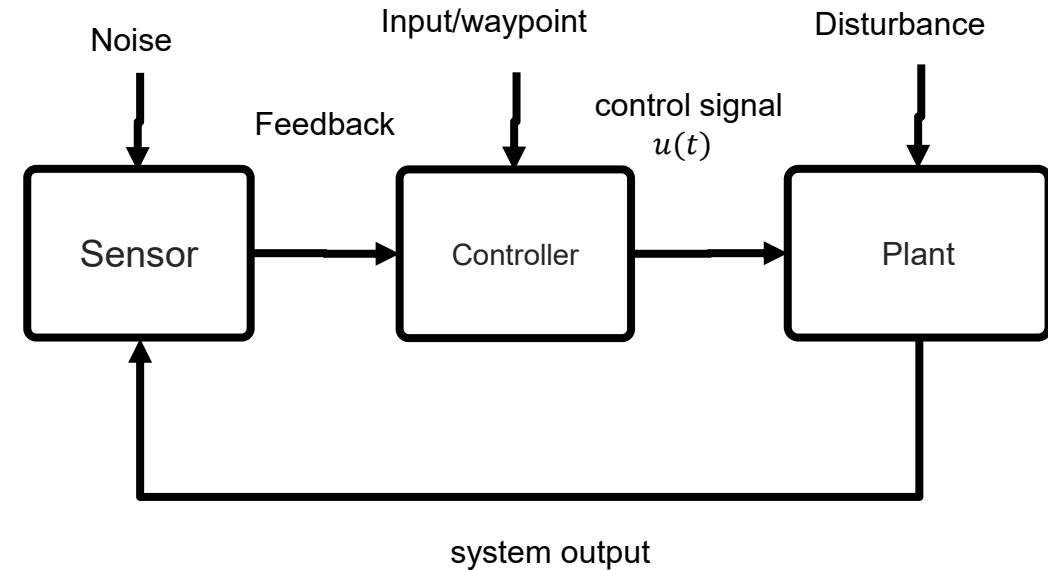
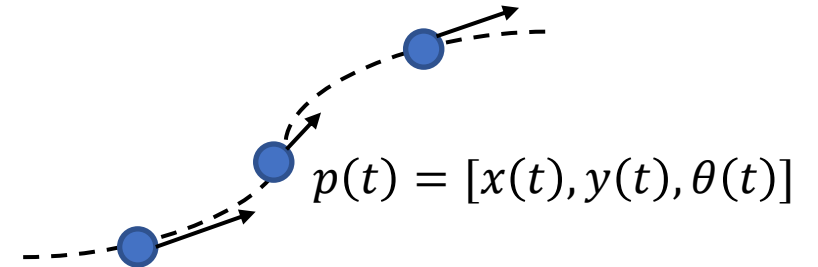
$v_B > 0$: forward motion; $\delta_B > 0$: left turn.

Path following control

The path is a parameterized curve. A time parameter gives a trajectory (including timing).

A higher-level planner computes the path (e.g., hybrid A*, RRT).

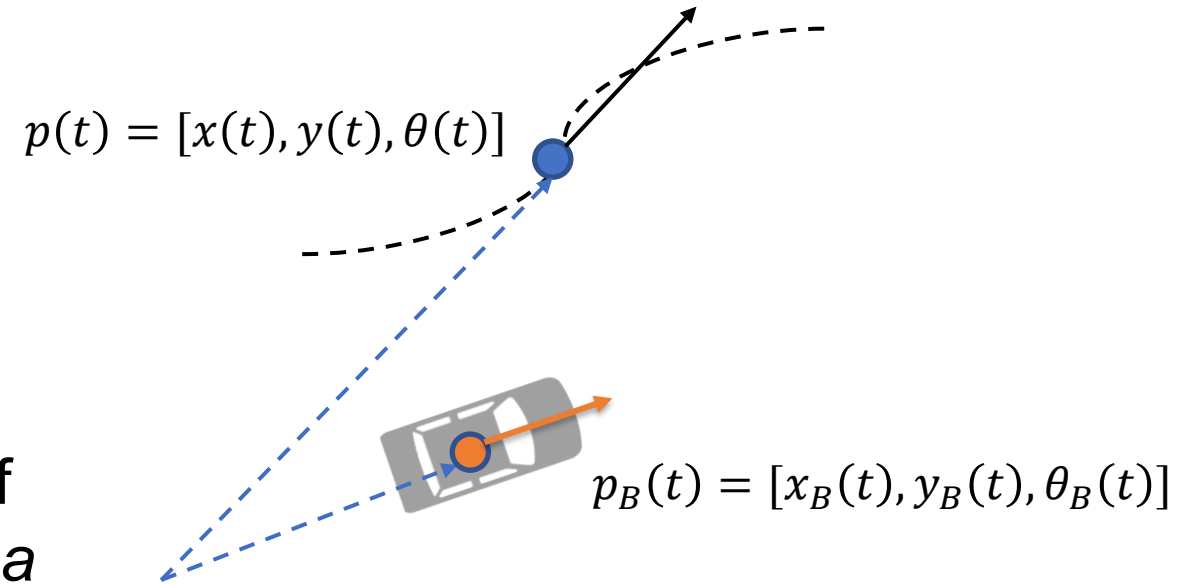
With a reference-point schedule, each point specifies a desired pose $p(t)$.



Path following control

Desired instantaneous pose $p(t)$

How to define error between actual pose $p_B(t)$ and desired pose $p(t)$ in the form of $x_d(t) - x(t)$ so that then we can develop a control law



Controller for bicycle model: find the errors

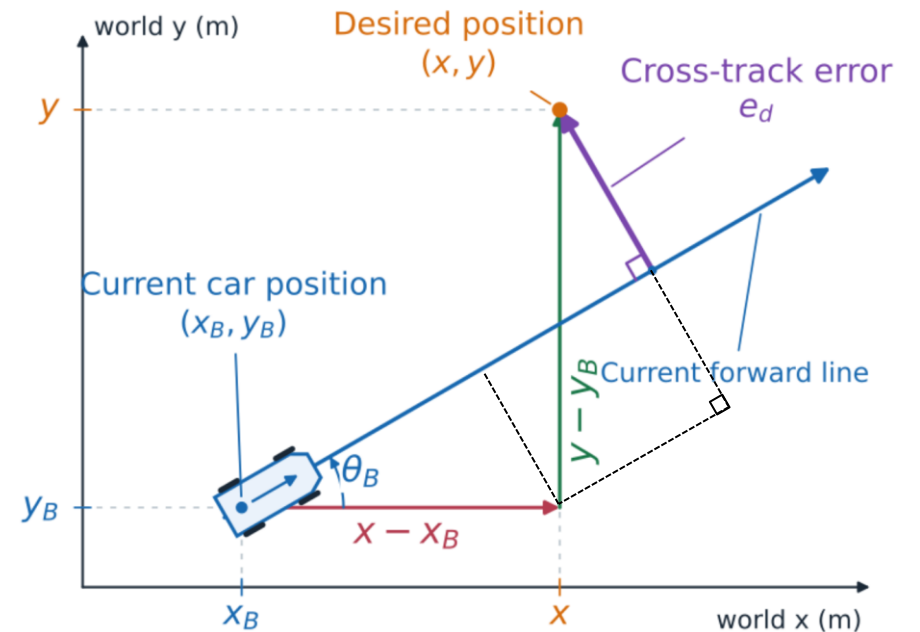
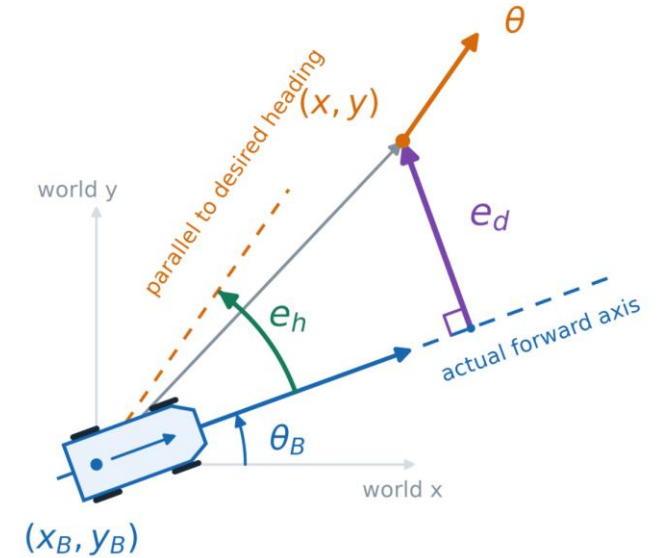
$$\text{Dynamics } \begin{bmatrix} \dot{x}_B \\ \dot{y}_B \\ \dot{\theta}_B \end{bmatrix} = \begin{bmatrix} v_B \cos \theta_B \\ v_B \sin \theta_B \\ \frac{v_B}{l} \tan \delta_B \end{bmatrix}$$

Heading error: $e_h = \text{wrap}(\theta - \theta_B)$ (wrap angles to $[-\pi, \pi]$)

Cross-track error: $e_d = -\sin \theta_B (x - x_B) + \cos \theta_B (y - y_B)$

(positive when the target is to the vehicle's left)

Combined error: $e = e_h + \alpha e_d, \alpha > 0$



Bang bang controller for bicycle model

$$\text{Dynamics } \begin{bmatrix} \dot{x}_B \\ \dot{y}_B \\ \dot{\theta}_B \end{bmatrix} = \begin{bmatrix} v_B \cos \theta_B \\ v_B \sin \theta_B \\ \frac{v_B}{l} \tan \delta_B \end{bmatrix}$$

Heading error: $e_h = \text{wrap}(\theta - \theta_B)$

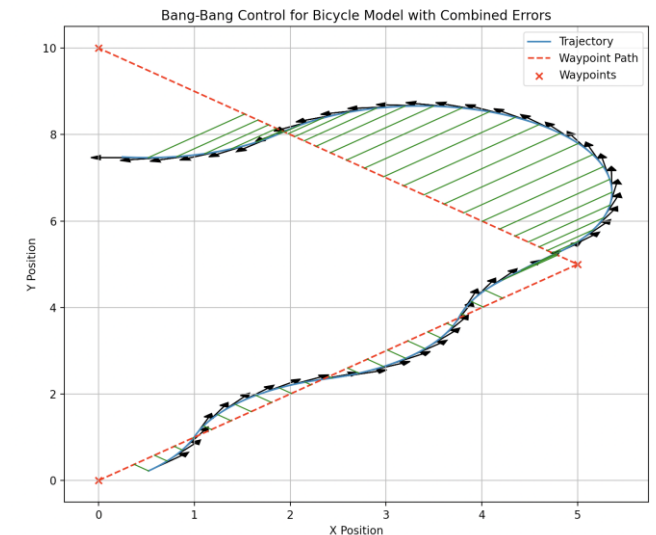
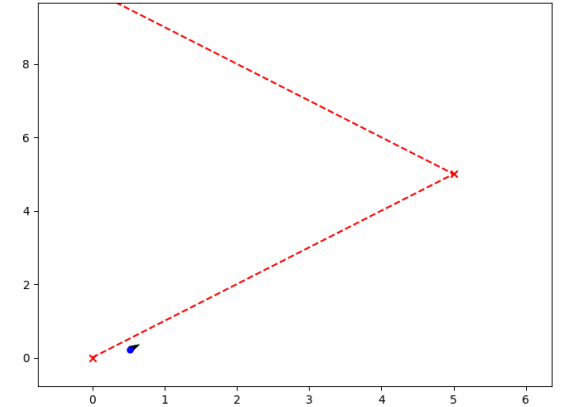
Cross-track error: $e_d = -\sin \theta_B (x - x_B) + \cos \theta_B (y - y_B)$
(positive when the target is to the vehicle's left)

Combined error: $e = e_h + \alpha e_d, \alpha > 0$

Bang-bang controller (straight-path example):

if $e > 0$: $\delta_B = \delta_{\max}$; if $e < 0$: $\delta_B = -\delta_{\max}$

if $e = 0$: $\delta_B = 0$; $0 < \delta_{\max} < \pi/2$



Proportional control for bicycle model

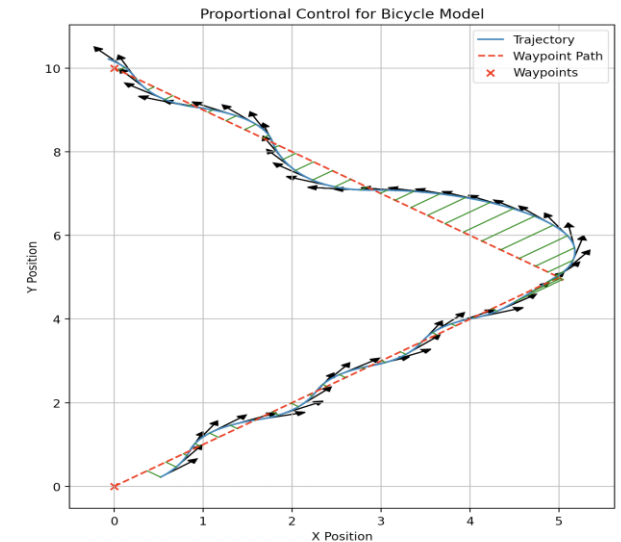
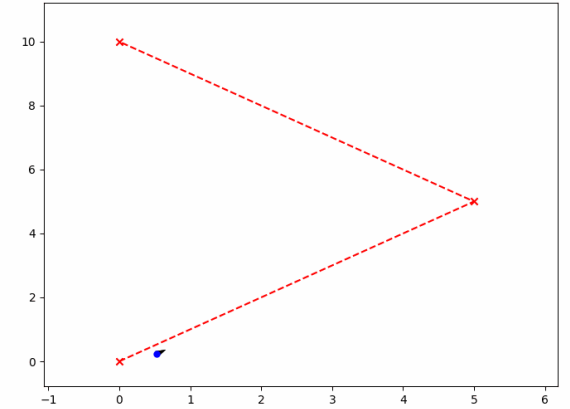
$$\text{Dynamics } \begin{bmatrix} \dot{x}_B \\ \dot{y}_B \\ \dot{\theta}_B \end{bmatrix} = \begin{bmatrix} v_B \cos \theta_B \\ v_B \sin \theta_B \\ \frac{v_B}{l} \tan \delta_B \end{bmatrix}$$

Heading error: $e_h = \text{wrap}(\theta - \theta_B)$

Cross-track error: $e_d = -\sin \theta_B (x - x_B) + \cos \theta_B (y - y_B)$

Proportional controller: $\delta_B = \text{clip}(K_h e_h + K_d e_d, -\delta_{\max}, \delta_{\max})$

$K_h > 0, K_d > 0$; finite steering limits.



Illustrative run; parameters unavailable.

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- Linear systems

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Linear dynamical system and solutions

A linear dynamical model is **linear jointly in state and input**. It covers a large number of real applications in control, and allows easy analysis.

$$\dot{x}(t) = A(t)x(t) + B(t)u(t)$$

$A(t) \in \mathbb{R}^{n \times n}$, $B(t) \in \mathbb{R}^{n \times m}$; they may vary with time.

For nonlinear dynamics, we can **linearize near an equilibrium** (x_e, u_e) :

$$A = \left. \frac{\partial f}{\partial x} \right|_{(x_e, u_e)}, B = \left. \frac{\partial f}{\partial u} \right|_{(x_e, u_e)}$$

For deviations $\eta = x - x_e$, $v = u - u_e$: $\dot{\eta} \approx A\eta + Bv$.

Linear systems allow efficient control design and analysis.

Given $x_0 \in \mathbb{R}^n$, $t_0 \in \mathbb{R}$, $u \in PC(\mathbb{R}, \mathbb{R}^m)$,

the solution is $\xi(\cdot, t_0, x_0, u): \mathbb{R} \rightarrow \mathbb{R}^n$.

Example: Simple linear model of an economy

Illustrative normalized model (not an economic forecast).

x : national income (state)

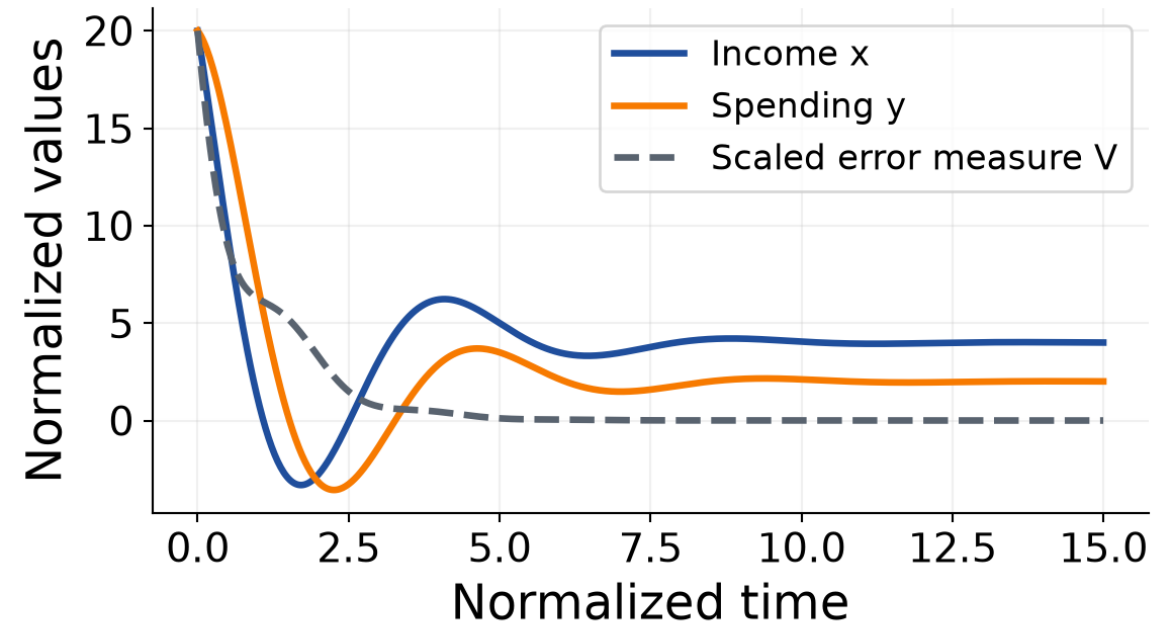
y : rate of consumer spending (state)

g : rate of government expenditure (input)

$$\dot{x} = x - \alpha y$$

$$\dot{y} = \beta(x - y - g)$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} 1 & -\alpha \\ \beta & -\beta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} 0 \\ \beta \end{bmatrix} g$$



Plot: $\alpha = \beta = 2, g = 2$; equilibrium (4,2).

Solutions of linear systems: continuity

$$\dot{x}(t) = A(t)x(t) + B(t)u(t) \quad \text{— Eq. (2)}$$

Assume continuous A, B and $u \in PC(\mathbb{R}, \mathbb{R}^m)$.

D_u : the input's jump times.

Theorem*. For any initial time t_0 , state x_0 and input u :

1. $\xi(\cdot, t_0, x_0, u)$ is continuous in time.

It is differentiable for every $t \notin D_u$.

2. For fixed t, t_0, u , the map $x_0 \mapsto \xi(t, t_0, x_0, u)$ is continuous.

The state does not jump when a finite input changes suddenly.

Solutions of linear systems: superposition

For the same model, any $a_1, a_2 \in \mathbb{R}$:

3. **Linearity / superposition** (combine initial states AND inputs):

$$\begin{aligned}\xi(t, t_0, a_1 x_{01} + a_2 x_{02}, a_1 u_1 + a_2 u_2) \\ = a_1 \xi(t, t_0, x_{01}, u_1) + a_2 \xi(t, t_0, x_{02}, u_2)\end{aligned}$$

4. **Decomposition**:

$$\xi(t, t_0, x_0, u) = \xi(t, t_0, x_0, 0) + \xi(t, t_0, 0, u)$$

With a fixed nonzero input, dependence on the initial state is generally affine.

Linear time invariant system: A, B do not change over time

$$\dot{x}(t) = Ax(t) + Bu(t)$$

Define matrix exponential:

$$\begin{aligned} e^{At} &= I + At + \frac{(At)^2}{2!} + \dots \\ &= \sum_{k=0}^{\infty} \frac{(At)^k}{k!} \end{aligned}$$

Theorem (solution of the LTI system):

$$\xi(t, t_0, x_0, u) = e^{A(t-t_0)}x_0 + \int_{t_0}^t e^{A(t-\tau)}Bu(\tau) d\tau$$

Zero input

Zero state

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Nineteen Dubious Ways to Compute the Exponential of a Matrix, Twenty-Five Years Later*

Cleve Moler[†]
Charles Van Loan[‡]

Abstract. In principle, the exponential of a matrix could be computed in many ways. Methods involving approximation theory, differential equations, the matrix eigenvalues, and the matrix characteristic polynomial have been proposed. In practice, consideration of computational stability and efficiency indicates that some of the methods are preferable to others, but that none are completely satisfactory. Most of this paper was originally published in 1978. An update, with a separate bibliography, describes a few recent developments.

Key words. matrix, exponential, roundoff error, truncation error, condition

AMS subject classifications. 15A15, 65F15, 65F30, 65L99

PII. S0036144502418010

Example

State x : position $x_1(0) = 0$ m, velocity $x_2(0) = -2$ m/s.

Input $u(t) = f_a(t) = 6$ N for $t \geq 0$.

$m = 3$ kg, $b = 0.6$ N·s/m, $k = 1.5$ N/m.

$$m \frac{dx_2(t)}{dt} = u(t) - b \frac{dx_1(t)}{dt} - kx_1(t)$$
$$x_2(t) = \frac{dx_1(t)}{dt}$$

